

COMBINATIONAL LOGIC DESIGN

Module.2

CDA 3103C

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Combinational logic

- Truth Tables, Logic Equations, and Gates
 - NOT, AND, OR, NAND, NOR, XOR, . . .
 - Minimal set
- Axioms and theorems of Boolean algebra
 - Proofs by re-writing
 - Proofs by perfect induction
- Gate logic
 - Networks of Boolean functions
 - Time behavior
- Canonical forms
 - Two-level
 - Incompletely specified functions
- Simplification
 - Boolean cubes and Karnaugh maps
 - Two-level simplification

Digital Systems

- Digital electronics operate with only two voltage levels of interest: a high voltage and a low voltage. All other voltage values are temporary and occur while transitioning between the values.



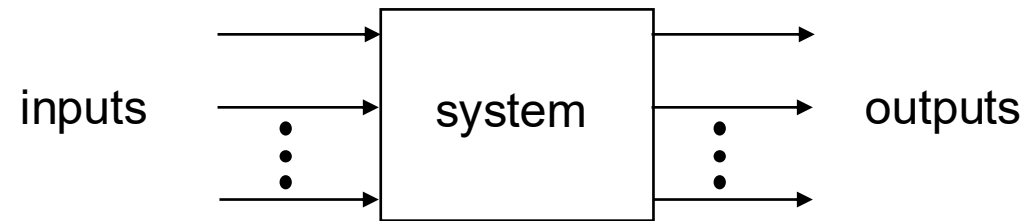
Digital Systems

- Low voltage is associated with 0s
- High voltage is associated with 1s
- The actual voltage values may differ from system to system



Logic Blocks and Block Diagrams

- Computational “blocks” perform a set of logical functions in either a combinational or sequential fashion.

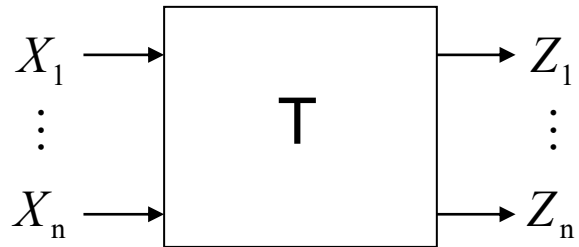


- A block diagram is a simple model of these systems that shows only the inputs and outputs.

Combinational vs. Sequential

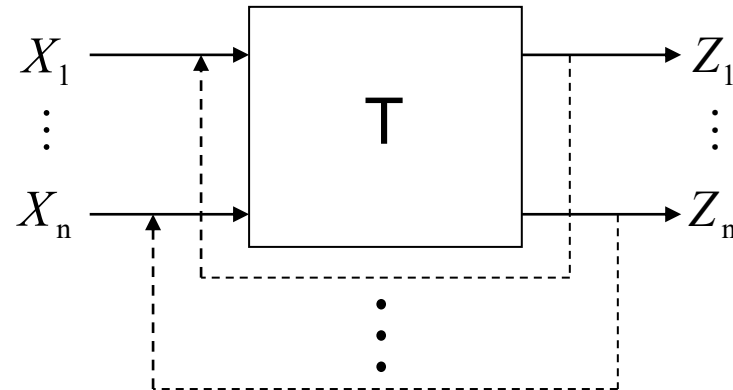
- Combinational:

- No Feedback
- Output defined completely in terms of the Inputs.



- Sequential:

- With feedback
- System goes through different states
- New state depends on Inputs and current state.

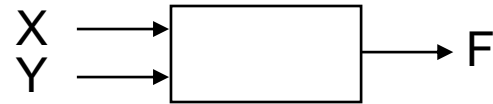


Truth Tables

- Combinational logic blocks can be completely specified by defining the output values for each possible set of input values.
- This is done using a truth table.
- For a logic block with n inputs, there are 2^n entries in the truth table. Each entry specifies the value of all the outputs for that particular input combination.

Possible Logic Functions

- If there are 2 input variables, there should be $2^2 = 4$ entries in the truth table.



- How many different functions of 2 input variables can we make?

Possible Logic Functions

X	Y	16 possible functions (F0–F15)															
0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
		X <u>and</u> Y		X		Y		X <u>xor</u> Y		X <u>or</u> Y		X <u>nor</u> Y <u>not</u> (X <u>or</u> Y)		X <u>xnor</u> Y		<u>not</u> Y	
																<u>not</u> X	
																X <u>nand</u> Y <u>not</u> (X <u>and</u> Y)	

- There are 16 possible functions of 2 input variables
- In general, there are 2^{2^n} functions of n inputs

Possible Logic Functions (Cont'd)

Boolean Function	Boolean Expression	Boolean Operator
NULL	$F = 0$	None
AND	$F = xy$	Dot (.)
Inhibition	$F = xy'$	x/y
Transfer/Buffer	$F = x$	None
Inhibition	$F = x'y$	y/x
Transfer/Buffer	$F = y$	None
Exclusive-OR	$F = xy' + x'y$	$x \oplus y$
OR	$F = x + y$	$x + y$
NOR	$F = (x + y)'$	$x \downarrow y$
Equivalence (Exclusive-NOR)	$F = xy + x'y'$	$(x \oplus y)'$
Complement/Inverter/NOT	$F = y'$	Y'
Implication	$F = x + y'$	$x \subset y$
Complement/Inverter/NOT	$F = x'$	x'
Implication	$F = x' + y$	$y \subset x$
NAND	$F = (xy)'$	$x \uparrow y$
Identity	$F = 1$	None

Truth Tables (Example)

- Consider a logic function with three inputs, A , B , and C , and three outputs, D , E , and F . The function is defined as follows:
 - D is true if at least one input is true,
 - E is true if exactly two inputs are true,
 - F is true only if all three inputs are true.

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Inputs			Outputs		
A	B	C	D	E	F
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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0	1	1	1		
1	0	0	1		
1	0	1	1		
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0	0	1	1	0	
0	1	0	1	0	
0	1	1	1	1	
1	0	0	1	0	
1	0	1	1	1	
1	1	0	1	1	
1	1	1	1	0	

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0	0	1	1	0	0
0	1	0	1	0	0
0	1	1	1	1	0
1	0	0	1	0	0
1	0	1	1	1	0
1	1	0	1	1	0
1	1	1	1	0	1

Boolean Algebra

- Deals with a set of variables (operands) combined with a set of operators.
 - Variables denoted by X, Y, Z , etc.
 - Variables take binary values:
Either “0” or “1” (“false” or “true”)
- Operators: **NOT**, **AND**, **OR**
- All logical operations can be described using these three operators.

OR

- The OR operator is written as $+$, as in $A + B$.

A	B	A + B (OR)
0	0	0
0	1	1
1	0	1
1	1	1

- The OR operation is also called a *logical sum*.

AND

- The AND operator is written as $*$, as in $A * B$.

A	B	$A * B$ (AND)
0	0	0
0	1	0
1	0	0
1	1	1

- The AND operation is also called a *logical product*.

NOT

- The unary operator NOT is written as A' or $\neg A$.

A	A'
0	1
1	0

Axioms and theorems of Boolean algebra

- Identity
 - $X + 0 = X$
 - $X \cdot 1 = X$
- Null
 - $X + 1 = 1$
 - $X \cdot 0 = 0$
- Idempotency:
 - $X + X = X$
 - $X \cdot X = X$
- Involution:
 - $(X')' = X$
- Inverse:
 - $X + X' = 1$
 - $X \cdot X' = 0$
- Commutative:
 - $X + Y = Y + X$
 - $X \cdot Y = Y \cdot X$
- Associativity:
 - $(X + Y) + Z = X + (Y + Z)$
 - $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$

Axioms and theorems of Boolean algebra

- Distributive:
 - $X \cdot (Y + Z) = (X \cdot Y) + (X \cdot Z)$
 - $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$
- Uniting:
 - $X \cdot Y + X \cdot Y' = X$
 - $(X + Y) \cdot (X + Y') = X$
- Absorption:
 - $X + X \cdot Y = X$
 - $X \cdot (X + Y) = X$
 - $(X + Y') \cdot Y = X \cdot Y$
 - $(X \cdot Y') + Y = X + Y$

Axioms and theorems of Boolean algebra

- de Morgan's:
 - $(X + Y + \dots)' = X' \cdot Y' \cdot \dots$
 - $(X \cdot Y \cdot \dots)' = X' + Y' + \dots$

Proving theorems with Rewriting

- Using the axioms of Boolean algebra:

- e.g., prove the uniting theorem:

	$X \cdot Y + X \cdot Y' = X$
Distributive law	$X \cdot Y + X \cdot Y' = X \cdot (Y + Y')$
Inverse law	$X \cdot (Y + Y') = X \cdot (1)$
Identity law	$X \cdot (1) = X \checkmark$

- e.g., prove the absorption theorem:

$$X + X \cdot Y = X$$

Identity law	$X + X \cdot Y$	$= X \cdot 1 + X \cdot Y$
Distributive law	$X \cdot 1 + X \cdot Y$	$= X \cdot (1 + Y)$
Null law	$X \cdot (1 + Y)$	$= X \cdot (1)$
Identity law	$X \cdot (1)$	$= X \checkmark$



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